PCA

03 May 2026 13:40

Principle component Analysis (PCA)

PCA is dimensionality-reduction technique that transforms the original features into a new set of of feature called principle components while retaining as much variance(info) as possible.

→ The new components are

(i) linear combinations of original variable.
(ii) Mutually uncorrelated
(iii) Ordered according to the amount of variance they capture.

PCA is an orthogonal transformation from potentially correlated axes to uncorrelated axes, where the new axes are priciple component.

EX:-

3 Original feature → 2 principal Component.

Core Steps of Principle Component Analysis:

Data → Mean-centring → Covariencematrix → Eigenvalues/Eigenvector → sort (descending order) create feature vector(select component) → Recast the data along the PCA.

EX -

Student	Hours studied (X_1)	MOCK Test Score (X_2)
S_1	2	4
S_2	4	5
S_3	6	7
S_4	8	7
S_4	10	12

① Representing data in matrix.

$$X = \begin{bmatrix} 2 & 4 \\ 4 & 5 \\ 6 & 7 \\ 8 & 7 \\ 10 & 12 \end{bmatrix}$$

Step1 : Mean-centring

$\bar{X}_1 = (2+4+6+8+10)/5$
$= 30/5 = 6$

$\bar{X}_2 = (4+5+7+7+12)/5$
$35/7 = 7$

Substracting these means from the centred matrix

$$B = \begin{bmatrix} 2-6 & 4-7 \\ 4-6 & 5-7 \\ 6-6 & 7-7 \\ 8-6 & 7-7 \\ 10-6 & 12-7 \end{bmatrix} = \begin{bmatrix} -4 & -3 \\ -2 & -2 \\ 0 & 0 \\ 2 & 0 \\ 4 & 5 \end{bmatrix}$$

(Step2) - Compute the Covarience Matrix

Let calculate variance of X_1

$$Var(X_1) = \frac{(-4)+(-2)^2+0^2+2^2+4^2}{4} = \frac{40}{4} = 10$$

$$Var(X_2) = \frac{(-3)^2+(-2)^2+0^2+0^2+5^2}{4} = \frac{38}{4} = 9.5$$

NOW:

$$Cov(X_1, X_2) = \frac{(-4)(-3)+(-2)(-2)+(0)(0)+(2)(0)+(4)(5)}{4} = \frac{36}{4} = 9$$

$$C = \begin{bmatrix} 10 & 9 \\ 9 & 9.5 \end{bmatrix}$$

(Step3) - Compute Eigenvalues & eigenvector

$\det(C - \lambda I) = 0$

$$\det\begin{bmatrix} 10-\lambda & 9 \\ 9 & 9.5-\lambda \end{bmatrix} = 0$$

$\Rightarrow (10-\lambda)(9.5-\lambda) - 81 = 0$
$\Rightarrow 95 - 10\lambda - 9.5\lambda + \lambda^2 - 81 = 0$
$\Rightarrow \lambda^2 - 19.5\lambda + 14 = 0$

$\lambda_1 = 18.75$
$\lambda_2 = 0.75$

$a=1, b=-19.5, c=14$

$$\lambda = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$\lambda = \frac{19.5 \pm \sqrt{(19.5)^2-4(1)(14)}}{2}$$

$$\lambda = \frac{19.5 \pm \sqrt{380.25-56}}{2} = \frac{19.5 \pm \sqrt{324.25}}{2}$$

$$\lambda = \frac{19.5 + 18.006}{2} = \frac{37.506}{2}$$

$\lambda_1 = 18.753$, $\lambda_2 = \frac{19.5 - 18.006}{2} =$

$\lambda_2 = \frac{1.494}{2} = 0.7465$

Step 3(b) finding Eigen vector

for $\lambda_1 = 19.75$

$$= \det\left(\begin{bmatrix} 10-18.75 & 9 \\ 9 & 9.5-18.75 \end{bmatrix}\right)\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$= \begin{bmatrix} -8.75 & 9 \\ 9 & -9.2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$= 8.75x_1 + 9x_2 = 0$
$\Rightarrow 9x_2 = 8.75x_1$
$\Rightarrow x_2 = \frac{8.75x_1}{9}$
$\Rightarrow x_1 = 1, \; x_2 = 0.972$

$$v_1 = \begin{bmatrix} 1 \\ 0.975 \end{bmatrix}$$

$$v_{1norm} = \frac{1}{1.395}, \frac{0.9725}{1.395}$$

$$v_{1norm} = \begin{bmatrix} 0.7169 \\ 0.6972 \end{bmatrix} \leftarrow PC_1$$

→ total variance covered by these values

$$PC_1 = \frac{\lambda_1}{\lambda_1 + \lambda_2} \times 100$$

$$= \frac{18.75}{18.75+0.75} \times 100$$

$$= \frac{18.75}{19.50} \times 100$$

$= 96.15\%$

It cover over 96% of raw changes in test patterns.

(Step4) creating Feature Vector

As λ_1 capture 96.15% of entire datasets variance we can discard the second principal component ($\lambda_2 = 0.75$)

So

$$v = \begin{bmatrix} 0.7169 \\ 0.6972 \end{bmatrix}$$

Step5: Recast the data Along the principal component

final Transform Data

$= B \times v$

$$= \begin{bmatrix} -4 & -3 \\ -2 & -2 \end{bmatrix}$$